Causal Explainable AI with Additive Noise Models: Pairwise Structure Discovery and Interventional Feature Attribution

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***Abstract*—Supervised learning models capture statistical associations, but standard feature attributions do not reveal how variables causally interact or how perturbations propagate through underlying data-generating mechanisms. This paper presents an end-to-end framework for causal explainable artificial intelligence (XAI) using Additive Noise Models (ANMs) on continuous observational data. The framework first infers a candidate causal structure through pairwise ANM evaluations. For each variable pair, directional asymmetry is evaluated using Gaussian Process Regression with cross-fitted residuals and the Hilbert-Schmidt Independence Criterion (HSIC). Low-confidence edges are filtered via percentile thresholding, and directed cycles are resolved deterministically by minimum-confidence edge elimination to guarantee a directed acyclic candidate graph (DAG). The resulting DAG parameterizes structural equations that enable interventional sampling, from which local Causal Feature Attribution (CAI) is computed by propagating interventions through downstream causal mechanisms to a target predictive model. The methodology is evaluated through controlled ground-truth synthetic benchmarks alongside continuous observational datasets and contextualized reference comparisons. On a five-node synthetic benchmark with known causal ground truth, the framework recovers two of four direct edges while identifying one transitive shortcut ($\mathbf{F1} = \mathbf{0.5714}$, $\mathbf{SHD} = \mathbf{3}$), exposing the structural limits of bivariate conditioning. Interventional attribution yields stable, statistically bounded feature rankings conditional on the candidate DAG, separating salient structural contributions from Monte-Carlo sampling variance. Furthermore, comparative evaluations against discrete adaptations, model-specification diagnostics, and variational architectures characterize the operational trade-offs between continuous nonparametric regression, model invalidation under confounding, and sample efficiency.**

***Index Terms*—Causal explainable AI, additive noise models, HSIC, causal discovery, causal feature attribution, directed acyclic graphs.**

I. Introduction

Modern machine learning models achieve high predictive accuracy on tabular and structured domains, yet predictive utility does not imply causal understanding. A feature may exhibit strong predictive importance because it is an immediate cause, a downstream consequence, or a proxy correlated through unobserved latent factors. Conventional post-hoc attribution methods evaluate model behavior under isolated feature perturbations, which can generate off-manifold samples and ignore structural dependencies among variables.

To address these limitations, causal explainable AI integrates structural causal models with post-hoc explanation. Under Additive Noise Model (ANM) assumptions, observational distributions often exhibit directional asymmetry: regressing an effect on its cause yields independent residual noise, whereas regressing the cause on the effect generally produces residuals dependent on the regressor [1]. Under the ANM assumptions, this asymmetry provides a basis for inferring causal direction from continuous observational data without combinatorial search over the full graph space.

However, translating bivariate causal asymmetries into an end-to-end explainability pipeline presents two fundamental methodological challenges: (1) how to assemble pairwise directional decisions into a coherent, topologically valid Directed Acyclic Graph (DAG) while managing weak associations, cycles, and transitive shortcuts; and (2) how to propagate local feature interventions through the learned structural mechanisms so that downstream predictive attributions reflect causal dependencies rather than isolated perturbations.

This paper proposes an integrated causal discovery and attribution framework based on nonlinear pairwise ANMs. The proposed method evaluates bivariate directions using Gaussian

Process (GP) regression with cross-fitted residuals and Hilbert-Schmidt Independence Criterion (HSIC) dependence tests. An empirical confidence margin filters weak edges, after which directed cycles are deterministically pruned to guarantee a directed acyclic candidate graph. Structural ANM equations fitted along the candidate DAG then support interventional sampling, enabling local Causal Feature Attribution (CAI) that explicitly propagates feature interventions through downstream causal mechanisms to a target predictive model.

Accordingly, the experimental evaluation addresses three primary research questions: (RQ1) Can pairwise nonlinear ANM analysis recover meaningful candidate causal structures from continuous observational data, and what are the structural consequences of bivariate conditioning? (RQ2) How does the proposed pairwise approach compare with representative discrete, diagnostic, and variational ANM methodologies under controlled ground truth and latent confounding? (RQ3) Can the inferred structural model support stable local feature attribution through interventional sampling with calibrated confidence bounds?

The primary contributions of this work are:

1) A pairwise nonlinear ANM causal discovery framework employing Gaussian Process regression, five-fold cross-fitted residuals, and HSIC independence scoring;
2) A graph construction procedure combining confidence-margin percentile thresholding, minimum-confidence cycle pruning, and formal topological DAG validation;
3) An interventional feature attribution mechanism (CAI) that propagates local feature clamps through structural equations to quantify causal influence on predictive outputs;
4) Methodological characterization and comparative analysis: uncovering the structural trade-offs of pairwise ANM discovery—specifically the emergence of transitive shortcuts, the behavior of relative asymmetry under latent confounding, and comparative trade-offs against discretized and variational reference paradigms [1], [2], [4].

II. Related Work

Additive Noise Models and Causal Discovery: The Additive Noise Model represents an effect variable as a nonlinear function of its causes plus an independent additive noise term. Peters et al. [1] demonstrated that ANM assumptions enable causal direction identification in discrete and integer domains by exploiting residual independence asymmetries. For continuous domains, nonlinear extensions using non-parametric regression and kernel independence tests have been widely investigated. To assess whether assumed ANMs reflect underlying data mechanisms, Schultheiss and Bühlmann [2] formulated an out-of-sample diagnostic for testing overall and partial causal well-specification using rank statistics and multi-split hypothesis aggregation.

Multivariate and Confounded ANMs: Scaling causal discovery beyond bivariate pairs requires addressing indirect paths and latent confounding. Parida et al. [3] proposed a multivariate complete conditioning ANM to eliminate transitive dependencies in full graphs. Addressing hidden variables, Qiao et al. [4] introduced Confounding Cascade ANM (CCANM), leveraging variational autoencoders to separate latent confounders from cascade noise. Computational scalability has also been explored through geometric coresets by Zhao et al. [5]. These approaches provide essential theoretical context regarding the challenges of multi-hop transitivity and latent confounding in structural estimation.

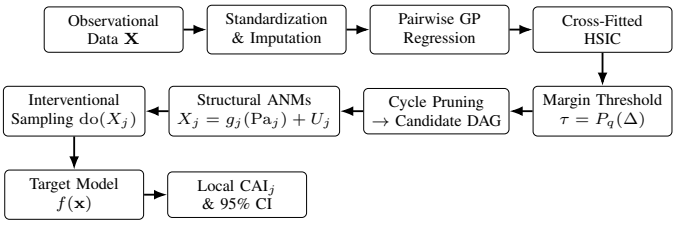


Fig. 1. Architecture of the proposed causal discovery and attribution framework. Observational data are processed through pairwise GP-HSIC evaluations, margin filtering, and cycle pruning to construct a candidate DAG, which drives structural interventional feature attribution.

Causal Feature Attribution: Whereas associative feature importance measures sensitivity to isolated inputs, causal attribution quantifies the expected change in model output under structural interventions $\mathrm{do}(X_j = x_j)$. By propagating perturbations along a candidate causal graph, causal attribution respects inter-feature relationships, avoiding evaluations on physically impossible configurations and isolating the downstream impact of causal mechanisms.

Research Gap: Existing ANM-based approaches address causal direction, model well-specification, multivariate conditioning, or latent confounding from distinct perspectives. However, a lightweight end-to-end framework that combines pairwise nonlinear ANM discovery with graph construction and intervention-based feature attribution remains comparatively underexplored. The proposed framework addresses this integration while explicitly evaluating the structural limitations introduced by pairwise discovery.

III. Proposed Method

A. Problem Formulation and Data Preprocessing

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote an observational dataset with n instances and p continuous features. Features are standardized to zero mean and unit variance. Preprocessing includes median imputation where necessary. Standardized features ensure uniform scale for kernel bandwidth estimation in Gaussian Process regression and independence testing.

B. Pairwise ANM Causal Direction Discovery

For each unordered pair of continuous variables (X, Y), the framework evaluates both directional hypotheses:

$$Y = f(X) + R_f, \tag{1}$$

$$X = g(Y) + R_r, \tag{2}$$

where f and g are estimated via Gaussian Process regression with Radial Basis Function (RBF) kernels and white noise kernels ($\alpha = 10^{-3}$). To mitigate optimistic bias from evaluating training residuals, five-fold cross-fitting is employed: residuals for each fold are generated strictly from out-of-fold predictions. In-sample residual evaluation on flexible nonparametric kernels can artificially deflate residual dependence by overfitting the regressor; out-of-fold cross-fitting decouples regressor fitting from independence scoring.

Residual independence is quantified using the Hilbert-Schmidt Independence Criterion (HSIC) with median Euclidean heuristic bandwidths. Let $h_f = \mathrm{HSIC}(X, R_f)$ and $h_r = \mathrm{HSIC}(Y, R_r)$. The inferred causal direction is selected as the one yielding lower residual dependence:

$$\widehat{X \to Y} = \begin{cases} X \to Y, & h_f < h_r, \\ Y \to X, & h_r \leq h_f. \end{cases} \tag{3}$$

The confidence margin associated with the orientation is defined as:

$$\Delta = |h_f - h_r|. \tag{4}$$

A larger margin indicates more pronounced residual asymmetry between the competing directions.

C. Graph Construction and DAG Validation

Pairwise evaluation across all $\binom{p}{2}$ pairs yields a directed multigraph that may contain weak associations, transitive edges, and directed cycles. The framework applies a two-stage sparsification and acyclicity procedure:

1) **Confidence-Margin Thresholding:** A percentile threshold $\tau = P_q(\{\Delta_k\}_{k=1}^m)$ is computed over all pairwise margins, where q is a configurable percentile (default $q = 75$). Edges with $\Delta < \tau$ are removed to suppress low-confidence orientations.
2) **Cycle Elimination:** If directed cycles exist in the filtered graph, cycles are detected via depth-first search, and the edge with the lowest confidence margin Δ in each cycle is pruned. Deterministic tie-breaking is applied when margins are identical.

The resulting graph is validated as a Directed Acyclic Graph, and a topological sorting $\pi = (\pi_1, \ldots, \pi_p)$ is established. Crucially, the resulting graph is interpreted as a candidate causal structure whose validity depends on the ANM assumptions and causal sufficiency, rather than as an automatically verified ground-truth causal DAG. Furthermore, because pairwise evaluations do not condition on intermediate variables, indirect ancestral paths (e.g., $X \to Z$ in a chain $X \to Y \to Z$) can satisfy residual independence and may be retained in candidate graphs alongside direct causal edges.

D. Causal Feature Attribution (CAI)

Once the candidate DAG $\mathcal{G}$ is established, structural ANMs are fitted along the graph topology. For each node X_j with parent set $\mathrm{Pa}(j)$, the structural mechanism is:

$$X_j = g_j(X_{\mathrm{Pa}(j)}) + U_j, \tag{5}$$

where g_j is fitted via Gaussian Process regression (with $g_j \equiv 0$ for root variables), and U_j represents exogenous residual noise modeled via fitted parametric Laplace distributions (fitted to marginal data for roots and to regression residuals for non-roots).

Given a target predictive model $f : \mathbb{R}^p \to \mathbb{R}$, local Causal Feature Attribution for feature j at observation $\mathbf{x}$ is defined as:

$$\mathrm{CAI}_j = \mathbb{E}[f(\mathbf{X}) \mid \mathrm{do}(X_j = x_j)] - \mathbb{E}[f(\mathbf{X})], \tag{6}$$

where $\mathbb{E}[f(\mathbf{X})]$ denotes the baseline model expectation. The interventional expectation is computed by drawing K Monte-Carlo samples: clamping $X_j = x_j$ across all draws and evaluating structural equations forward in topological order π, where root variables are drawn from their fitted marginal Laplace distributions and non-intervened endogenous variables evaluate $g_k(X_{\mathrm{Pa}(k)}) + U_k$ using upstream simulated values. The baseline expectation $\mathbb{E}[f(\mathbf{X})]$ is evaluated analogously without clamping. For classification models, f denotes the predicted probability of the selected class; for regression models, f denotes the scalar prediction.

Monte-Carlo sampling with K draws is employed to compute standard errors:

$$\mathrm{SE}_j = \sqrt{\frac{\mathrm{Var}(f_{\mathrm{do},j})}{K} + \frac{\mathrm{Var}(f_{\mathrm{base}})}{K}}, \tag{7}$$

yielding 95% confidence intervals $\mathrm{CAI}_j \pm 1.96\,\mathrm{SE}_j$. Normalized attribution is given by $\mathrm{nCAI}_j = \mathrm{CAI}_j / (\sum_k |\mathrm{CAI}_k| + 10^{-8})$.

IV. Experimental Setup

A. Experimental Design and Benchmark Objectives

The empirical evaluation investigates three distinct objectives: (1) evaluating structural recovery, precision-recall tradeoffs, and transitive shortcuts against controlled ground truth; (2) evaluating behavior under latent confounding when causal sufficiency is violated; and (3) assessing dimensional scalability, candidate DAG density ($p \in [5, 30]$), and interventional attribution stability on continuous observational datasets where true causal graphs are unknown.

B. Synthetic Ground-Truth and Confounding Protocols

To evaluate causal discovery against objective ground truth, a continuous synthetic SCM is generated with five variables:

$$X_1 \to X_2 \to X_3 \to X_4 \to X_5, \tag{8}$$

with $N = 300$ samples, nonlinear mechanisms ($\tanh$, $\sin$, polynomials), and independent Laplace exogenous noise ($\mathrm{random_state} = 42$). The true edge set contains four directed edges: $\{(X_1, X_2), (X_2, X_3), (X_3, X_4), (X_4, X_5)\}$. Performance is measured using True Positives (TP), False Positives (FP, partitioned into transitive ancestral paths and spurious connections), False Negatives (FN), Precision, Recall, F1 score, Structural Hamming Distance (SHD), and wall-clock runtime.

TABLE I
DATASET SUMMARY AND INFERRED CANDIDATE DAG STATISTICS

Dataset	Obs.	Feat.	Pairs	Retained Edges	Density	Cycles
Wine	178	13	78	20 ($\tau = 0.002919$)	0.256	0
Breast Cancer	569	30	435	109 ($\tau = 0.004371$)	0.251	0
Airfoil Noise	1503	6	15	4 ($\tau = 0.002647$)	0.133	0
Synthetic SCM	300	5	10	3 ($\tau = 0.001753$)	0.300	0

TABLE II
LOCAL CAUSAL FEATURE ATTRIBUTION FOR WINE OBSERVATION 0

Feature	CAI	Normalized CAI	95% Confidence Interval
proline	0.542078	0.375056	[0.512411, 0.571745]
malic_acid	0.210111	0.145373	[0.174077, 0.246145]
alcohol	0.195086	0.134977	[0.159631, 0.230540]
flavanoids	0.175165	0.121194	[0.139274, 0.211057]
od280/od315	0.127793	0.088419	[0.092274, 0.163313]
alcalinity_of_ash	0.084320	0.058340	[0.048876, 0.119764]
color_intensity	0.037062	0.025643	[0.001504, 0.072621]
nonflavanoid_phenols*	0.030004	0.020759	[-0.005315, 0.065323]
magnesium*	0.019440	0.013450	[-0.015833, 0.054713]
total_phenols*	0.012186	0.008432	[-0.023014, 0.047387]
proanthocyanins*	0.005927	0.004101	[-0.029152, 0.041007]
ash*	0.004586	0.003173	[-0.030398, 0.039569]
hue*	-0.001565	-0.001083	[-0.036569, 0.033439]

* Confidence intervals cross zero, indicating estimates within Monte-Carlo variance.

To evaluate behavior under hidden variables, a synthetic latent confounding cascade is implemented following Qiao et al. [4], where an unobserved confounder C simultaneously influences both X and Y alongside a causal path $X \to Y$ ($N = 300$).

C. Observational Continuous Benchmarks

Three observational datasets provide diverse dimensional and sample regimes: the Wine dataset (178 samples, 13 features), the Breast Cancer Wisconsin Diagnostic (WDBC) benchmark (569 samples, 30 features, 435 pairs), and the Airfoil Self-Noise dataset (1503 samples, 6 features). Because observational datasets lack verified causal mechanisms, their inferred graphs serve as descriptive candidate topologies rather than ground-truth evaluations. For the Wine dataset, a Logistic Regression model ($P(\text{class } 0)$) provides a black-box predictive interface $f(\mathbf{x})$ for interventional attribution. In predictive diagnostics, stratified five-fold cross-validation confirms task learnability ($98.33\% \pm 1.36\%$ accuracy), serving as a diagnostic check of model utility rather than an internal component of CAI.

D. Reference-Method Comparison Protocol

The proposed method is compared against three established reference methodologies under transparent adaptations:

1) **Peters-Style Discrete ANM [1]:** Formulated originally for discrete variables, this method is adapted to continuous data via deterministic quantile binning ($K = 5$ default, with sensitivity evaluated across $K \in \{3, 5, 7, 10\}$), followed by mode regression and χ^2 independence tests.
2) **Schultheiss & Bühlmann Directional Diagnostic [2]:** Designed as an ANM model-specification diagnostic, this approach is adapted into an experimental directional edge orienter by comparing multi-split aggregated p-values between competing directions.
3) **Comparable PyTorch CCANM (Qiao et al.) [4]:** A variational autoencoder architecture incorporating latent confounders and cascade noise. The implementation optimizes ELBO with margin $\delta = 0.05$. It is evaluated under the common $N = 300$ benchmark constraint, acknowledging that the original method targets large sample regimes ($N \geq 2500$).

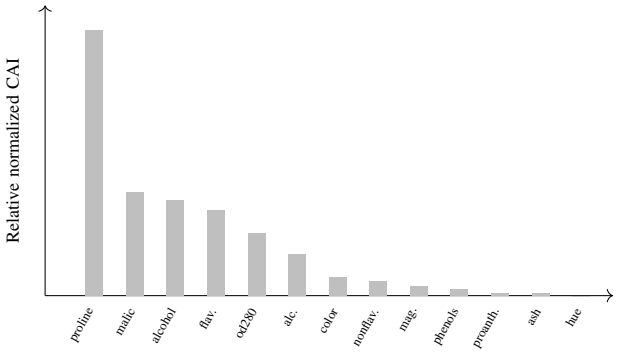


Fig. 2. Relative normalized CAI ranking on Wine observation 0. Bar heights reflect relative magnitude; numerical values are reported in Table II.

Two additional reference methods—the multivariate ANM of Parida et al. [3] and the coreset ANM of Zhao et al. [5]—are retained as literature references but were not experimentally reproduced due to paywalled source material. All experiments use fixed random seeds and deterministic evaluation settings where applicable. The synthetic benchmark uses $N = 300$ and random seed 42. The benchmark generation, evaluation procedures, and automated validation tests are maintained with the study artifacts.

V. RESULTS AND DISCUSSION

A. Structural Recovery and Ground-Truth Benchmark Comparison

Table III reports the quantitative comparison on the synthetic ground-truth benchmark ($X_1 \to \cdots \to X_5$, $N = 300$). The empirical results reveal distinct structural behaviors across the four methodologies:

1) **Proposed Method:** Recovers two direct edges ($X_1 \to X_2$ and $X_2 \to X_3$) and misses two edges ($X_3 \to X_4$ and $X_4 \to X_5$). It incurs one false positive: the transitive edge $X_2 \to X_4$ (an indirect ancestral path). It introduces zero spurious connections, achieving Precision $= 0.6667$, Recall $= 0.5000$, F1 $= 0.5714$, SHD $= 3$, and runtime 1.22 s.
2) **Peters-Style Discrete ANM Adaptation:** Recovers zero direct edges (TP $= 0$, F1 $= 0.0$). As shown in Table IV, varying the binning parameter $K \in \{3, 5, 7, 10\}$ produces either an empty graph ($K = 3$) or transitive/spurious edges ($K \in \{5, 7, 10\}$). Quantile discretization collapses smooth non-linear variations into

TABLE III
SYNTHETIC BENCHMARK COMPARISON ($X_1 \to X_2 \to X_3 \to X_4 \to X_5$, $N = 300$)

Method	TP	FP	FN	Precision	Recall	F1	SHD	Runtime
Proposed method	2	1	2	0.6667	0.5000	0.5714	3	1.22 s
Peters et al. (2011)*	0	1	4	0.0000	0.0000	0.0000	5	1.20 s
Schultheiss & Bühlmann (2024)**	3	5	1	0.3750	0.7500	0.5000	6	4.69 s
Qiao et al. (2021)***	0	3	4	0.0000	0.0000	0.0000	6	1.69 s

*Evaluated through continuous-data quantile-binning adaptation.
**Evaluated under experimental directional well-specification adaptation.
***Evaluated using comparable PyTorch CCANM implementation under common $N = 300$ benchmark.

TABLE IV
PETERS DISCRETIZATION SENSITIVITY ACROSS BIN PARAMETER K

K	Discovered Edges	TP	FP (Trans / Spur)	F1	Interpretation
3	$\emptyset$	0	0 (0 / 0)	0.0000	Empty graph (underfitting)
5	$X_1 \to X_4$	0	1 (1 / 0)	0.0000	Transitive edge
7	$X_1 \to X_4$, $X_1 \to X_3$	0	2 (2 / 0)	0.0000	Transitive edges
10	4 edges†	0	4 (2 / 2)	0.0000	2 transitive + 2 spurious

†Edges at $K = 10$: $\{(X_4, X_1), (X_1, X_3), (X_2, X_5), (X_5, X_1)\}$. Zero TP persisted across all K on this continuous benchmark.

step functions, demonstrating that discrete ANM adaptations exhibit limited fidelity on continuous manifolds.

3) **Schultheiss & Bühlmann Directional Adaptation:** Achieves the highest recall (0.7500, recovering 3 direct edges) but admits 5 false positive edges. Notably, all five false positives are transitive ancestral paths ($X_1 \to X_3$, $X_1 \to X_4$, $X_1 \to X_5$, $X_2 \to X_4$, $X_2 \to X_5$) with zero spurious edges, yielding Precision $= 0.3750$, F1 $= 0.5000$, and SHD $= 6$ in 4.69 s.
4) **Comparable PyTorch CCANM (Qiao et al.):** Discovers three edges ($X_1 \to X_5$ transitive, $X_3 \to X_2$ reversed, $X_4 \to X_2$ spurious), achieving TP $= 0$, F1 $= 0.0$, and SHD $= 6$. On small sample sizes ($N = 300$), the deep latent confounder and cascade representations provide sufficient flexibility to fit competing directions with comparable ELBO values, reflecting sensitivity to sample size relative to the authors' large-scale domain ($N \geq 2500$).

B. The Transitive-Edge Dilemma in Pairwise Discovery

The benchmark results clarify an inherent property of pairwise causal discovery: the retention of transitive edges. In the synthetic chain SCM, the proposed method retained the indirect edge $X_2 \to X_4$. Applying the cross-fitted well-specification diagnostic of Schultheiss and Bühlmann [2] to this edge yielded $p = 0.200 > 0.05$, confirming that the fitted model $X_4 = \hat{f}(X_2) + E$ satisfies residual independence.

Crucially, statistical well-specification must not be conflated with direct causal parenthood. In the true SCM, X_2 is an ancestor of X_4 mediated through X_3 ($X_2 \to X_3 \to X_4$). Because compound noise along ancestral paths often remains approximately independent of the cause, pairwise ANMs cannot distinguish direct from indirect causes without conditional independence testing or aggressive thresholding.

TABLE V
EVALUATION UNDER LATENT CONFOUNDING CASCADE

Method	Outcome	Diagnostic Evidence	Interpretation
Proposed method	$X \to Y$	HSIC: 0.000627 vs. 0.005887	Relative residual asymmetry
Peters et al. [1]	Rejected	Both $p \ll 0.01$ (bad model fit)	ANM model incompatibility
Schultheiss [2]	$X \to Y$	Forward $p = 0.2800$ vs. 0.0000	Directional p-value selection
Qiao et al. [4]	Non-ident.	$\lvert\Delta\mathcal{L}\rvert = 0.0145 < \delta = 0.05$	Deliberate abstention

TABLE VI
DESCRIPTIVE COMPARISON ON OBSERVATIONAL REAL DATASETS

Dataset	Method	Edges	Density	Cycles	Runtime
Airfoil ($N = 1503$)	Proposed method	4	0.133	0	5.01 s
Airfoil ($N = 1503$)	Peters et al. [1]	1	0.033	0	1.29 s
Airfoil ($N = 1503$)	Qiao et al. [4]	8	0.267	0	1.47 s
Wine Sub. ($N = 178$)	Proposed method	3	0.150	0	0.56 s
Wine Sub. ($N = 178$)	Peters et al. [1]	1	0.050	0	1.31 s
Wine Sub. ($N = 178$)	Qiao et al. [4]	6	0.300	0	0.58 s

No causal ground truth is available for these datasets; results reflect descriptive topology and wall-clock runtime.

C. Behavior Under Latent Confounding and Causal Sufficiency

Table V reports outcomes under the synthetic latent confounding setup ($X \to Y$ with unobserved C, $N = 300$). When causal sufficiency is violated, the evaluated paradigms exhibit contrasting methodological responses: *1) Proposed Method:* Orients $X \to Y$ based on relative residual asymmetry (forward HSIC 0.000627 vs. reverse 0.005887), demonstrating that relative asymmetry alone cannot distinguish direct causation from unobserved confounding. *2) Peters et al.:* Rejects both directions as "bad model fit" ($p_{\text{fwd}} \sim 10^{-9}, p_{\text{bwd}} \sim 10^{-21}$), reflecting appropriate rejection when confounding violates standard ANM assumptions. *3) CCANM (Qiao et al.):* The ELBO margin falls below the decision threshold ($\lvert\Delta\mathcal{L}\rvert = 0.0145 < \delta = 0.05$), abstaining by classifying the pair as non-identifiable. These outcomes illustrate an essential methodological distinction: relative asymmetry heuristics force edge orientations under assumed sufficiency, whereas hypothesis testing and margin thresholds enable conservative model invalidation and abstention.

D. High-Dimensional Observational Topology and Computational Scalability

Tables I and VI summarize topological characteristics and runtimes across observational datasets. Because observational data lack verified causal graphs, these experiments evaluate candidate graph formation and computational scaling rather than causal correctness. On Wine ($p = 13$), the 75th-percentile margin threshold ($\tau = 0.002919$) retains 20 candidate directed edges with zero cycles (density 0.256). On the 30-feature Breast Cancer dataset (435 pairs), the framework retains 109 directed edges ($\tau = 0.004371$, density 0.251) with zero cycles, confirming that confidence-margin pruning scales to higher dimensions. On Airfoil ($p = 6$), 4 edges are retained (density 0.133).

Regarding runtime and density trade-offs (Table VI), discrete ANM yields sparse graphs (density 0.033–0.050, runtimes 1.29–1.31 s); CCANM yields denser graphs (density

0.267–0.300, runtimes 0.58–1.47 s); and the proposed method yields intermediate density (0.133–0.150, runtimes 0.56–5.01 s). Theoretically, the pairwise discovery phase has time complexity $\mathcal{O}(p^2 n^3)$, reflecting $\binom{p}{2}$ pairwise evaluations requiring five-fold cross-fitted Gaussian Process regressions with $\mathcal{O}(n^3)$ kernel inversions. Observed runtimes mirror these scaling dynamics (Table VI), remaining tractable for moderately sized tabular datasets ($n \leq 2000, p \leq 30$).

E. Interventional Causal Feature Attribution and Stability

Table II and Fig. 2 report local CAI values for Wine observation 0 (baseline output $P(\text{class } 0) = 0.311357$ under $K = 1000$). Proline produces the highest causal attribution (CAI $= 0.542078$, normalized attribution 0.375056), followed by *malic_acid*, *alcohol*, and *flavanoids*. The top seven features exhibit 95% confidence intervals strictly excluding zero, confirming that their positive attribution is distinct from Monte-Carlo sampling variance under the fitted model (Fig. 2). These intervals quantify simulation variance for the estimated interventional expectation, distinct from epistemic uncertainty about the candidate causal graph itself.

Robustness analysis comparing $K = 1000$ and $K = 2000$ Monte-Carlo samples yields baseline values of 0.311357 and 0.305126, respectively. The Spearman rank correlation between allocations is $\rho = 0.9066$, and the ordering of the top seven features is identically preserved, verifying numerical stability. However, stable attribution rankings reflect consistent numerical estimation under the fitted model rather than proving real-world causal correctness; causal attribution remains conditional on candidate DAG fidelity.

VI. Limitations

The proposed framework is subject to several methodological limitations:

1) **Bivariate Scope and Transitive Ancestral Paths:** Evaluating variable pairs in isolation without conditioning on intermediate variables allows indirect ancestral dependencies to satisfy residual independence, retaining transitive shortcuts ($X \to Z$ in $X \to Y \to Z$).
2) **Causal Sufficiency and Latent Confounder Blindness:** The framework assumes causal sufficiency (no unobserved common causes). Under latent confounding, relative residual asymmetry can force an edge orientation even when both directions violate ANM independence assumptions.
3) **Heuristic Sparsification Threshold:** The percentile margin cutoff $\tau = P_q(\Delta)$ is an empirical heuristic lacking formal asymptotic false discovery rate (FDR) control, requiring domain-specific calibration.
4) **Reference Adaptation Constraints:** Quantile discretization degrades continuous non-linear resolution for Peters-style ANMs; directional orienting via Schultheiss & Bühlmann p-values is an experimental adaptation of a specification test; and the small-sample benchmark ($N = 300$) does not reflect CCANM's large-sample design regime ($N \geq 2500$). Literature references [3], [5] were not reproduced experimentally due to paywalled source access.
5) **Absence of Observational Ground Truth:** Real-world datasets (Wine, WDBC, Airfoil) lack known causal graphs; empirical topologies must remain descriptive.
6) **Local Explanation Scope and Model Dependency:** CAI provides local instance-level explanations conditioned on the fitted candidate structural equations rather than global invariant causal mechanisms.
7) **Computational Complexity:** Exact Gaussian Process regression requires $\mathcal{O}(n^3)$ computation and $\mathcal{O}(n^2)$ memory per pairwise evaluation, limiting scalability to large datasets without sparse or coreset approximations.

VII. Conclusion and Future Work

This paper presented an integrated methodology for causal explainable artificial intelligence combining pairwise Additive Noise Model discovery with interventional feature attribution. Directional asymmetries evaluated via Gaussian Process regression with five-fold cross-fitted residuals and HSIC independence scoring yield directed acyclic candidate structures via percentile margin filtering and deterministic cycle pruning, parameterizing downstream structural equations for interventional sampling.

The experimental evaluation substantiates key methodological conclusions: (1) pairwise continuous ANMs successfully recover non-linear directional mechanisms against controlled ground truth ($\text{F1} = 0.5714$, $\text{SHD} = 3$), but exhibit an intrinsic susceptibility to transitive shortcut edges; (2) interventional propagation yields stable, statistically bounded attribution rankings that separate structural influence from sampling variance; and (3) contextualized comparisons demonstrate the structural limits of discrete adaptations and the necessity of conservative model invalidation under latent confounding.

Future work will investigate: (1) hybrid conditioning schemes to prune transitive shortcuts without combinatorial graph search; (2) adaptive thresholding based on false discovery rate control; and (3) coreset approximations to scale continuous Gaussian Process regressions to larger sample sizes.

References

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